Pipes and Cisterns (Subset of Time & Work)

↳ Leak (−ve work done)

Q21. Two pipes can fill the cistern in 10 hours and 12 hours respectively, while the third empties it in 20 hours. If all the pipes are opened simultaneously, then the cistern will be filled in

A) 9 hours B) 8.5 hours C) 8 hours D) 7.5 hours ✓

$E_{P_1} = \frac{1}{10}$ $E_{P_2} = \frac{1}{12}$ $E_3 = -\frac{1}{20}$

$\frac{1}{10} + \frac{1}{12} + -\frac{1}{20} = \frac{6+5-3}{60} = \frac{8}{60}$

$T = \frac{60}{8} = 7.5$

Q24. Two pipes can fill a tank in 20 and 24 minutes respectively and a waste pipe can empty 3 gallons per minute. All the three pipes working together can fill the tank in 15 minutes. The capacity of the tank in gallons is

A) 100 B) 110 C) 120 ✓ D) 140

$E_1 = \frac{1}{20}$ $E_2 = \frac{1}{24}$

$\frac{1}{20} + \frac{1}{24} + E_w = \frac{1}{15}$

$E_w = \frac{1}{15} - \frac{1}{20} - \frac{1}{24} = \frac{8-6-5}{120} = \frac{-3}{120} = \frac{-1}{40}$

$Cap = 3 \times 40 = 120$

Q22. A cistern is normally filled in 5 hours. However, it takes 6 hours when there is a leak in its bottom. If the cistern is full, in what time can the leak empty half of it?

A) 6h B) 5h C) ~~30h~~ D) 15h ✓

$E_T = \frac{1}{5}$ $E_T + E_L = \frac{1}{6}$

$E_L = \frac{1}{6} - \frac{1}{5} = \frac{-1}{30}$

Q25. Three taps P, Q and R can fill a tank in 12 hours, 15 hours and 20 hours respectively. If P is open all the time and Q and R are open for one hour each alternately, starting with Q, then the tank will be full in how many hours?

A) 9 hours B) 7 hours C) 13 hours D) 11 hours

$E_P = \frac{1}{12}$ $E_Q = \frac{1}{15}$ $E_R = \frac{1}{20}$

$W_{1st} = E_P + E_Q = \frac{1}{12} + \frac{1}{15} = \frac{5+4}{60} = \frac{9}{60}$

$W_{2nd} = E_P + E_R = \frac{1}{12} + \frac{1}{20} = \frac{5+3}{60} = \frac{8}{60}$

$W_2 = \frac{9}{60} + \frac{8}{60} = \frac{17}{60}$ $W_6 = \frac{17}{60} \times 3 = \frac{51}{60} < 1$

$W_4 = \frac{17}{60} \times 2 = \frac{34}{60}$ $W_8 = \frac{17}{60} \times 4 = \frac{68}{60} > 1$